A Statistical Examination of Equity in Graduation Rates

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April 29, 2024

1 Introduction

One of the most crucial aspects of a healthy community is an educated populace[5]. A society where members are able to dissect knowledge and discern subtle nuances from various sources of information is immensely advantageous [6]. Of course, this does not necessitate that every person needs to pursue higher education and receive advanced degrees, as it can be seen as that a functioning society can be comprised of members who do not have an extensive education. Over the past hundred years, the United States and California have decided the minimum threshold for education is a high school diploma. This is the highest required level of schooling dependent adolescents must attend within the state. It follows then that United States is known to have the highest population of high school students[17].

Throughout history, low graduation rates have had deep lasting consequences on the government and economic growth, so it is critical that we determine what factors impact graduation rates the most [12]. Empirically, it is clear that graduation rates are affected by many demographic factors that lead to inequality within the United States [20]. As a state in which the diverse landscapes and populations are arguably its biggest advantage, California is not exempt from symptoms of inequality and injustice in the education system [11]. The size of the state along with the huge number of students causes many of California's school districts to have various techniques in administering statewide curriculum to best suit their students' needs[7]. While it would be simple to try to implement the methods provided by higher-performing districts, that would not provide the entire picture of why that specific district is successful in creating higher graduation rates. For that reason, there is a distinct need for deep analysis of graduation data and trends in order to determine how the educational systems can best support the students within the state of California.

The COVID-19 pandemic has had an untold effect on the majority of people's lives. This includes drastic consequences to the economy, environment, and other sectors of the global community [15]. Furthermore, many of these effects were seen within California and the overall nation, and some of the hardest hit institutions were the schools and education systems. This caused each state and even district to have to find other approaches for teaching [8]. Due to health and safety concerns, almost all students had to attend virtual classes during the beginning of the pandemic, and strict social distancing rules were in place during the proceeding school years [22]. Subsequently, California schools experienced a "learning lag" where the overall cohort of students had lower test scores nearing the end of 2021 compared to previous years. These results yet again highlighted the urgent need to adjust curriculum in the education system for accessibility and equity [14]. The Pandemic may have altered many factors of an individual's graduation chances. It can be argued that counties possessing a financial advantage have done significantly better throughout the COVID-19 pandemic due to their availability of resources outside of the school district [8]. Furthermore, the learning lag for the disadvantaged grew even more due to circumstances created during the COVID-19 pandemic, and a teacher's ability to be versatile in technology is increased dramatically [14]. Therefore, the pandemic has indicated a need for improvement on creating an equitable space for disadvantaged students.

Acknowledging the various methods that each of California's districts may employ, we examined the impact that demographic factors have on graduation rates such as student gender, ethnicity, socioeconomic disadvantages, teacher experience, and median household incomes in each California County [13]. Ideally, we hoped to find that demographic factors had no significant impact on graduation rates, but that was never assumed to be the case. Consequently, our statistical study constructed a predictive model based on how a student with differing factors was likely to succeed in high school graduation and the weight with which each factor should be addressed in terms of future education reform measures. We understood our research would not provide a solution to this unfortunate circumstance, but we were hopeful that future policymakers and educational administration could use our findings to improve equity for all Californians.

2 Methods

2.1 2019 Graduation Data

The majority of our data collection came directly from the California Department of Education database. Specifically, we pulled the data from the Certified Staff Experience database alongside the 4-Year Adjusted Cohort Graduation Rate database focused on district specific data in. We also drew data from the National Institute for Health to determine the median incomes of households in each county [13]. The data pulled from the National Institute of Health was an average from 2017-2021. Our chosen variables of study are listed alongside their abbreviations which can be seen throughout the paper (Table 1). From this data, we wanted to determine which predictor variables had the strongest

Category Class	Predictor Variable	Abbreviation
Gender	Female	GF
Gender	Male	GM
Race/Ethnicity	Asian	RA
Race/Ethnicity	African-American	RB
Race/Ethnicity	Unreported	RD
Race/Ethnicity	Filipino	RF
Race/Ethnicity	Hispanic	RH
Race/Ethnicity	Native American	RI
Race/Ethnicity	Pacific Islander	RP
Race/Ethnicity	Two or more Race/Ethnicities	RT
Race/Ethnicity	White	RW
County	Median Household Income	MHI
Staff	Average Teaching Years of Service	TAYS

Table 1: Demographic Variables and Abbreviations

correlation with graduation within the state using RStudio which allowed us to easily replicate the model with data from other academic school years. Due to the existence of teaching experience data, we chose the 2018-19 academic school year for our research. Logistic regression allowed us to calculate the logistical coefficient that determines the probability of graduation based on each predictor variable [18]. The logit equation is given by:

$$logit(G) = \beta_0 + \sum_{i=1}^{k} \beta_i X_i$$

For our research, $logit(G)$ represented the probability of graduating, β_i represented the coefficients of the predictor variables, and x_i represented the values of the predictor variables. The coefficients for our research were calculated in RStudio with the underlying statistical distribution being a binomial plot [16].

Unfortunately, our data did not include individual reports of each student and instead included the numerical count of students in each reporting category along with their respective graduation rates by district. Consequently, our model was unable to explore the intersections between demographic factors such as race and gender. Therefore, we built multiple logistic regression models with specific predictor variables that allowed for the most intersectionality. Because of the impact that teacher income had on student achievements and learning, we began with an inspection on the correlation between county median household income and the average years of experience teachers have in each district [10]. Our first logistic regression model set tested graduation rates with gender, median household income, teaching experience, and the interactions between these variables (Table 2).

Models	Formula
G_1	Null
G_2	Gender
G_3	MHI
G_4	Gender + MHI
G_5	TAYS
G_6	Gender + TAYS
G_7	Gender + MHI + TAYS
G_8	Gender + Gender:MHI
G_9	Gender + Gender:TAYS
G_{10}	Gender + MHI + TAYS + Gender:MHI + Gender:TAYS

Table 2: Gender Models

We then constructed a second set of logistic regression models which tested graduation rates in correlation with race, median household income, teaching experience and interactions (Table 3). We will also test the graduation rates against each district with no specific reporting category selected, median household income, and teaching experience. Our results provided $\beta-$coefficients for each variable creating an accurate prediction model. Where Teacher Average Years in Service and Median Household Income variables were included, we used their respective numerical values for our data.

Model Names	Formula
R_1	Null
R_2	Ethnicity
R_3	MHI
R_4	Ethnicity + MHI
R_5	TAYS
R_6	Ethnicity + TAYS
R_7	Ethnicity + MHI + TAYS
R_8	Ethnicity + Ethnicity:MHI
R_9	Ethnicity + Ethnicity:TAYS
R_{10}	Ethnicity + MHI + TAYS + Ethnicity:MHI + Gender:TAYS

Table 3: Race/Ethnicity Models

2.2 Model Analysis

Further analysis of our models created in the study was conducted with the calculation of the Akaike Information Criterion (AIC) where a lower score illustrates lower complexity within the specific model [2]. This allowed us to choose the best fitted model in terms of the data through the calculation of the weight w of each model using their respective AIC scores [21]. we calculated the weight with the following equation:

$$w_i = \frac{e^{-\frac{1}{2}\Delta_i(AIC)}}{\sum_{i=1}^{k} e^{-\frac{1}{2}\Delta_i(AIC}}$$

where $\sum w_i = 1$ for all of the models. This allowed us to select the final models providing the most accurate impacts that the demographic graduation statistics had on graduation rates.

2.3 COVID and School Year Variance

From the California Department of Education database, we used the 4-Year Adjusted Cohort Graduation Rate data of other academic school years to assess the effects of the COVID-19 pandemic on high school graduation rates. We classified each of the selected school years as Pre-COVID, Peak-COVID, and Post-COVID Eras [19]. With each data frames, we created a cumulative academic school year data frame that ran from the 2018 to 2023 school years. Our predictor variables for this logistic regression model were therefore the academic school years themselves (Table 4).

Category Class	Predictor Variable	Abbreviation
Academic Year	2017-2018	Pre
Academic Year	2018-2019	Pre
Academic Year	2019-2020	Peak
Academic Year	2020-2021	Peak
Academic Year	2021-2022	Post
Academic Year	2022-2023	Post

Table 4: Yearly Variables and their Abbreviations

Using these variables, we constructed binomial logistic regression models comparing graduation rates of each era separately along with different combinations of the years (Table 5). The teaching experience and median household income data were not be included within these models since they were not calculated with each individual year past the 2018-19 school year, and therefore were unable to provide an accurate depiction of their impact on yearly graduation rates.

Model Names	Formula
C_1	Null
C_2	Pre Vs. Peak Vs. Post
C_3	Pre Vs Peak+Post
C_4	Peak Vs. Pre+Post
C_5	Post Vs.Pre+Peak

Table 5: COVID-19 Models

3 Results

The creation of a box plot of the 2019 data gave us a preliminary view of how our variables correlated with graduation in all of California (Figure 1). It can be seen that students recorded as Asian or

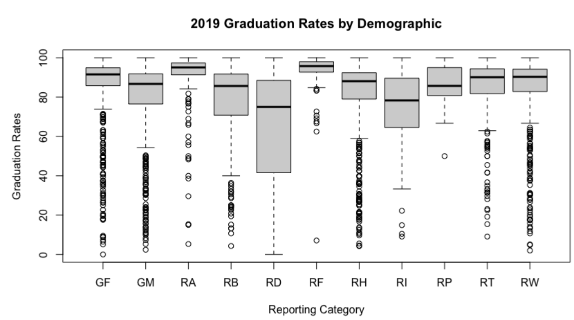

Figure 1: 2019 Graduation Rates by Demographic

Filipino had the highest mean graduation rates at 95.1% and 95.7% respectively while also having the lowest variance in comparison to all other reporting categories. Female students possessed an average graduation of 91.5% in comparison to 86.7% for their male counterparts. The greatest variance in graduation success was for students with Non Reported race or ethnicity with mean 75.0%, and Indigenous students had the second lowest mean at 78.3%. From our data set, we constructed a histogram showing the density of graduation rates of all students regardless of demographic within California (Figure 2). The overall frequency of numerical graduation rates taken at the district level within California showed graduation rates were right-skewed in density, and the mean graduation rate for all students was at 80.76%.

It can be seen that students recorded as Asian or Filipino had a higher graduation rate while also having a lower variance in their respective graduation success in comparison to all other reporting cat-

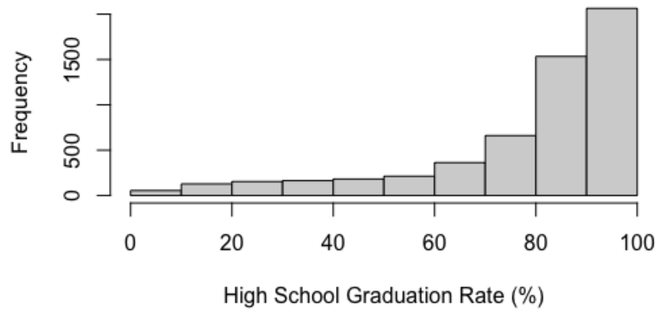

Figure 2: 2018-2019 Graduation Rates

egories. Notable low graduation rates were seen for reporting categories where students were reported as possessing a disability, reported as foster children, reported as homeless, and reported as English learning. The greatest variance in graduation success was present for students with an undetermined or unreported race or ethnicity. While not contributing to the creation of a predictive model, it was important to see what the actual graduation rates by reporting category were before the creation of a predictive regression model. The models showed that graduation rates and student gender had a strong impact with teaching experience within the districts and county median income. Male students were linked with lower graduation rates while also experiencing higher benefits from increased teaching experience in comparison to their female counterparts. For gender, the model incorporating median household income, average teaching years of service, and gender interaction's with these two variables G_{10} had the lowest AIC with Akaike weight $w = 0.818$. The next best fitting model G_7 had Akaike weight $w = 0.182$ (Table 5). All other models had far greater AIC scores, and therefore had Akaike weights $< 10^{-6}$ giving far worse models for the data [2].

Model	AIC	ΔAIC	w_i
G_{10}	14401	0	0.818
G_7	14404	3	0.182
G_9	14428	27	$< 10^{-6}$
G_6	14432	31	$< 10^{-6}$
G_4	15457	1056	$< 10^{-6}$
G_8	15459	1058	$< 10^{-6}$
G_2	15472	1071	$< 10^{-6}$
G_5	72317	57916	$< 10^{-6}$
G_3	76990	62589	$< 10^{-6}$
G_1	77002	62601	$< 10^{-6}$

Table 6: AIC of Gender Models and Weights w_i

The model G_{10} had far more information of influences and probability of graduation in comparison to model G_7 due to the inclusion of interactions between variables. All models provided β-coefficients alongside z-values used in Wald statistics (Appendix). The intercept of the G_{10} model was students labelled as female with $\beta_0 = -0.615$ while male students coefficient was $\beta_{GM} = -0.159$. Median household income was given $\beta_{MHI} = 6.29 \times 10^{-7}$ and teaching experience was given $\beta_{TAYS} = 3.68 \times 10^{-2}$. The impacts of household income varied by student gender as $\beta_{MHI} + \beta_{GF:MHI} = 5.14 \times 10^{-7}$ and $\beta_{MHI} + \beta_{GM:MHI} = 5.27 \times 10^{-7}$. Average teaching experience was also slightly more impactful for male students with β-coefficients $\beta_{TAYS} + \beta_{GF:TAYS} = 3.13 \times 10^{-2}$ and $\beta_{TAYS} + \beta_{GM:TAYS} = 3.19 \times 10^{-2}$.

For our race and ethnicity models, the data again suggested that graduation rates were impacted by students races and ethnicities. Asian and Filipino students had higher graduation rates in comparison to non-reporting and indigenous students. For race/ethnicity, the model incorporating median household income, average teaching years of service, and student race/ethnicity interactions with these variables was R_{10} which had the lowest AIC with Akaike weight $w = 0.1.00$. All other models had ΔAIC greater than 10 and were given an Akaike weight $< 10^{-6}$ (Table 7). The model R_{10} used the re-

Model	AIC	ΔAIC	w_i
R_{10}	19469	0	1.000
R_9	19594	125	$< 10^{-6}$
R_6	19770	301	$< 10^{-6}$
R_7	19772	303	$< 10^{-6}$
R_8	20494	1025	$< 10^{-6}$
R_2	20674	1205	$< 10^{-6}$
R_4	20675	1206	$< 10^{-6}$
R_5	72317	52848	$< 10^{-6}$
R_3	76990	57521	$< 10^{-6}$
R_1	77002	57533	$< 10^{-6}$

Table 7: AIC of Race/Ethnicity Models and Weights w_i

porting category for Asian students as its intercept with $\beta_0 = -0.212$ (Appendix). The only reporting category with a positive coefficient was Filipino students with $\beta_{RF} = 0.170$. Median household income and teaching experience had coefficients $\beta_{MHI} = 1.07 \times 10^{-6}$ and $\beta_{TAYS} = 2.75 \times 10^{-2}$. Median household income had the greatest positive impact on Non Reporting students graduation rates as $\beta_{MHI} + \beta_{RD:MHI} = 6.14 \times 10^{-6}$ respectively. Filipino, Hispanic, and Indigenous students graduation rates were negatively impacted by higher median household income according to the model. This negative impact was highest amongst Indigenous students with $\beta_{MHI} + \beta_{RI:MHI} = 2.67 \times 10^{-6}$. The model showed clear distinctions on the impacts of teaching experience for each reported race/ethnicity. Pacific Islander and Asian students were negatively impacted by greater teaching experience according to the model. Pacific Islanders had a slightly lower coefficient in comparison to their Asian counterparts at $\beta_{TAYS} + \beta_{RP:TAYS} = 2.80 \times 10^{-3}$. Again, Non-reporting students had the highest positive impact from higher teaching experience with $\beta_{TAYS} + \beta_{RD:TAYS} = 1.15 \times 10^{-1}$. The other coefficients aligned with our assumptions of the impacts race/ethnicity have on graduation rates.

Comparing the COVID-19 eras, the data suggests that graduation rates have experienced an increasing trend that was unaffected by the COVID pandemic. The best fitted model comparing the three eras C_2 had the lowest AIC with Akaike weight $w = 0.998$ (Table 8). The next model with ΔAIC around 10 was C_3 with Akaike weight $w = 0.002$ and can be mainly ignored when making predictions using the best fitted model. Again, all other models were much farther away in AIC scores and can therefore be ignored with weights $< 10^{-6}$. Thus, we choose to focus on the model C_2. All era's within the model were considered highly significant in showing correlation with changes of graduation rates (Appendix). The intercept of model C_2 was the Peak-COVID era with a coefficient $\beta_0 = -0.192$, Pre-COVID had the coefficient $\beta_{Pre} = -0.004$, and Post-COVID had the coefficient $\beta_{Post} = 0.029$.

Model	AIC	ΔAIC	w_i
C_2	435590	0	0.998
C_5	435603	13	0.002
C_3	436358	768	$< 10^{-6}$
C_4	436593	1003	$< 10^{-6}$
C_1	436784	1194	$< 10^{-6}$

Table 8: AIC of COVID Era Models and Weights w_i

4 Discussion

There are many factors which can impact graduation rates ranging from the students' home lives and the curriculum methods they learn under. There are significant outliers in school districts where each demographic factor have perfect near perfect graduation rates. Alternatively, there are also outlier districts in which an extreme low percentage of students graduate from each reporting category. We modelled these graduation rates by demographic categories to determine the general impact each group faces when attempting to graduate high school. We also provide some possible interaction with average teaching experience and median household income and their effects on each demographic of students.

Consequently, we see that higher pay in a county and greater experience for teachers within the district increase the graduation rates of most students regardless of gender. This increase in pay potentially leads to higher quality teaching and resources which would attract more experienced teachers to take and hold these positions [9]. Students are more likely to benefit within these environments [4]. In regards to race and ethnicity, students identified as Pacific Islanders are the only demographic that are slightly negatively impacted by higher teaching experience according to the model. They have one of the highest graduation rates among all reported demographics which could result in this negative fit with teaching experience due to high performance regardless. Increasing median household income has a minor negative impact on Filipino students' graduation rates which can also be explained with a similar line of reasoning.

It is surprising to see the negative impact increasing household income has on Hispanic and Indigenous students as suggested by the model. These two groups perform notably worse in comparison to their Asian and Filipino counterparts which would lead to the assumption that higher incomes would benefit their graduations, but this does not seem to be the case. Students belonging to the Unreported classification had significant positive impacts when increasing median household income and teaching experience, but this could be do to the category having a far smaller population sample size leading. Despite these discrepancies and outliers, we can still note the clear benefits of increasing funding in education and graduation rates for most students. Even by increasing median household income in counties, there are positive impacts on district funding and teacher's pay which creates greater flexibility for districts to create equity programs to assist certain groups to succeed. Teachers with an increase in certifications and knowledge lead to greater students success [3]. This could be a way to make progress for higher equity amongst all students within the state[4].

The COVID-19 pandemic has had drastic effects on the education system throughout California and the country. The pandemic altered the process of how curriculum was taught and led to drastic teacher retirement and fiscal restructuring [18]. We were shocked to not see any decrease in graduation rates during the height of the pandemic for these reasons. Student graduation rates slightly increased during the pandemic and showed a dramatic increase after during the 2022 school year. This could be explained by federal action taken to assist school systems financially during that time period [1]. Our model only examined the total graduation rates combining all reporting demographics per district, and future studies could examine if the yearly graduation rates impacted differently for each reporting category.

The largest obstacle that we encountered in the creation of our models was the collection and cleaning of the large data frames. Thus, the number of predictor variables included within the model is somewhat limited, and can be expanded upon in future study. Our models succeed in illuminating the distinctions of graduation rates amongst various student demographic groups. These models use simple statistical analysis to highlight these inequalities allowing for future studies into why these valuable results.

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Appendix: Coefficient Tables

β-coefficient	Estimate	Std. Error	z value	Pr(>|z|)
GF (Intercept)	-0.615	0.02	-24.67	0.00
GM	-0.158	0.04	-4.50	0.00
MHI	6.29×10^{-7}	0.00	4.31	0.00
TAYS	3.68×10^{-2}	0.00	24.83	0.00
GF:MHI	-1.15×10^{-7}	0.00	-0.56	0.58
GM:MHI	-1.02×10^{-7}	0.00	-0.32	0.58
GF:TAYS	-5.49×10^{-3}	0.00	-2.61	0.01
GM:TAYS	-4.89×10^{-3}	0.00	-1.52	0.01

Table 9: Model G_{10} Coefficients and z Values.

β-coefficient	Estimate	Std. Error	z value	Pr(>|z|)
GF (Intercept)	-0.659	0.02	-37.12	0.00
GM	-7.19×10^{-2}	0.00	-17.09	0.00
MHI	5.71×10^{-7}	0.00	5.50	0.00
TAYS	3.41×10^{-2}	0.00	32.41	0.00

Table 10: Model G_7 Coefficients and z Values.

	Estimate	Std. Error	z value	Pr(>|z|)
RA (Intercept)	-0.212	0.07	-3.18	0.00
RB	-0.893	0.11	-8.41	0.00
RD	-2.57	0.18	-14.48	0.00
RF	0.170	0.13	1.27	0.20
RH	-0.383	0.07	-5.38	0.00
RI	-0.685	0.31	-2.24	0.03
RP	0.148	0.36	0.41	0.68
RT	-0.604	0.13	-4.76	0.00
RW	-0.387	0.08	-5.14	0.00
MHI	1.07×10^{-6}	0.00	5.02	0.00
TAYS	2.75×10^{-2}	0.00	12.99	0.00
RA:MHI	-5.71×10^{-7}	0.00	-1.60	0.11
RB:MHI	4.23×10^{-7}	0.00	0.73	0.46
RD:MHI	5.07×10^{-6}	0.00	4.83	0.00
RF:MHI	-1.48×10^{-6}	0.00	-2.45	0.01
RH:MHI	-2.28×10^{-6}	0.00	-8.62	0.00
RI:MHI	-3.74×10^{-6}	0.00	-1.76	0.08
RP:MHI	2.05×10^{-6}	0.00	-1.22	0.22
RT:MHI	8.83×10^{-7}	0.00	1.37	0.17
RW:MHI	6.92×10^{-7}	0.00	1.41	0.15
RA:TAYS	-2.00×10^{-2}	0.00	-4.64	0.00
RB:TAYS	2.49×10^{-2}	0.00	5.05	0.00
RD:TAYS	8.88×10^{-2}	0.01	7.23	0.00
RF:TAYS	2.58×10^{-2}	0.01	-3.81	0.00
RH:TAYS	7.84×10^{-3}	0.00	3.05	0.00
RI:TAYS	2.91×10^{-2}	0.02	1.54	0.12
RP:TAYS	-2.85×10^{-2}	0.02	-1.41	0.16
RT:TAYS	7.28×10^{-3}	0.01	1.10	0.27
RW:TAYS	6.77×10^{-3}	0.01	1.38	0.22

Table 11: Model R_{10} Coefficients and z Values.

	Estimate	Std. Error	z value	Pr(>|z|)
Peak (Intercept)	-0.192	0.00	-257.43	0.00
Post	0.0290	0.00	27.75	0.00
Pre	-0.004	0.00	-3.89	0.00

Table 12: Model C_2 Coefficients and z Values.